

Introduction to Theoretical Linguistics

Phonetics, Phonology and Morphology

Assignment 7

Name _____ ID _____

Ex 1 Basic terminology

Which of the following best describes the relationship between a morpheme and an allomorph?

- A) A morpheme is a superficial phonetic realization, while an allomorph is an abstract mental unit.
- B) An allomorph is one of the alternative phonetic forms of a single morpheme, determined by its phonological environment.
- C) Allomorphs are different definitions or meanings that can be assigned to a single morpheme.
- D) Morphemes only appear in written language, whereas allomorphs only exist in spoken language.

Ex 2 Comparing hypotheses

The English past tense suffix is realized as distinct allomorphs, such as [t], [d], and [əd]. To explain how these forms surface, we must determine the single underlying form (the mental representation) stored in the speaker's lexicon, and the rules that derive the other variants. Consider the three competing hypotheses below, each proposing a different underlying form for the past tense morpheme:

Hypothesis A: The underlying form is / t /.

Hypothesis B: The underlying form is / d /.

Hypothesis C: The underlying form is / əd /.

Question 1: We prefer the hypothesis that explains the data using the **fewest, most natural, and most cross-linguistically common rules** (the principle of economy/simplicity). Under this view, which hypothesis is most likely to be adopted?

Question 2: If so, what additional phonological rules are to be recognized to yield the words *baked*, *used*, and *ended*?

Ex 3 Necessity of Morphology

Fill in the blanks with the correct verbs that should go into A and B.

A student argues:

“Why do we need the concept of ‘allomorphs’? When sounds change based on their environment, it’s just a mechanical sound change like assimilation. For example, we learned that [t], [d] and [əd] are the allomorphs of the same past tense morpheme, but we can predict which allomorph to use based on the preceding phonological environment. If so, it’s just phonology.”

A well-observed, point-taken comment, but we do have a reason. Admittedly, the distinction becomes much clearer in some languages than in English, but even with English data, we must learn the concepts of allomorph and morpheme. To see why, look at the data set below.

According to the phonological rules from Exercise 2, a verb ending in the voiceless sound [k] should automatically take the past tense allomorph [t]. This rule works perfectly for the verb *bake* → *baked* [beɪkt]. However, the verb ^(A) _____ also ends in the exact same sound, but its past tense surfaces as ^(B) _____. Because the exact same phonological environment produces two completely different outputs, we **cannot** explain English past tense formation using phonology alone. Therefore, the human brain must store these exceptions as distinct **allomorphs** tied to specific words.

Verb A: _____

Verb B: _____